Inductive Design of Maturity Models: Applying the Rasch Algorithm for Design Science Research

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Abstract. Maturity models are an established means to systematically document and guide the development of organizations using archetypal capability levels. Often, these models lack a sound foundation and/or are derived on the basis of an arbitrary design method. In order to foster the design of relevant and rigorous artifacts, this paper presents a method for maturity model construction that applies the Rasch algorithm and cluster analysis as a sound methodical foundation. The Rasch algorithm is widely used to improve scholarly intelligence and attainment tests. In order to demonstrate the application of the proposed method and to evaluate its usability and applicability, we present a design exemplar in the business intelligence domain.

Keywords: design science, maturity models, Rasch algorithm, business intelligence.

1 Introduction

In the field of Information Systems (IS) and Management Science, maturity models (MM) are steadily being applied both as an informed approach for continuous improvement [1, 2] and as a means for self or third party assessment [3, 4]. Since the introduction of the concept in the 1970s [e.g. 5, 6], a multiplicity of different MMs has been developed in academia and practice, making it an excellent example for design research [7]. Becker et al. [8] report that more than 1,000 articles refer to MMs and Mettler et al. [9] found more than 100 MMs.

Nonetheless, the concept has not been undisputed by criticism. For instance, Pfeffer & Sutton [10] argue that the purpose of MMs is to identify a gap, which can then be closed by subsequent improvement actions. However, lots of these models do not describe how decision makers effectively have to perform these actions. Hence, often a 'falsified certainty of success' is purported. Mettler and Rohner (2009) excoriated that yet today situational factors (e.g. corporate culture, organization structure, company size) are often left behind in favor of reducing complexity (thus assuming a static and somewhat simplistic world view). In the business intelligence

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(BI) domain, a recent methodical state-of-the-art analysis reveals that MMs often lack a theoretical foundation, are not well documented, and especially dismiss methodical requirements [11]. This is in line with Biberoglou & Haddad [12], who state that there exists no common and widely accepted design methodology and formal theoretical basis for MM construction so far.

The goal of this paper is to address the core issue outlined by Lahrmann et al. [11] and Biberoglou & Haddad [12]. Hence, the paper proposes a psychometric-driven approach for designing relevant *and* rigorous MMs based on the Rasch algorithm and cluster analysis. Advancing the knowledge base of artifact construction, this paper aims at contributing to the meta research of design science [13].

Following a basic design science process of "build" and "evaluate" [14, 15], the paper is structured as follows. In section 2, we outline the methodical foundations as well as theoretical considerations for the construction of MMs. In section 3, the underlying assumptions and our algorithm-based MM construction methodology are discussed in detail. In order to clarify and evaluate the suggested construction method, we develop an exemplary BI MM in section 4.1 Finally, we summarize our findings and offer suggestions for future work.

2 Foundations

2.1 Definition of Maturity and Demand for a Maturity Model

In general, 'maturity' can be defined as "the state of being complete, perfect or ready" [16]. Maturity thus implies an evolutionary progress in the demonstration of a specific ability or in the accomplishment of a target from an initial to a desired or normally occurring end stage. In the IS discipline, 'maturity' is rather regarded as "a measure to evaluate the capabilities of an organization" [17]. Following Becker et al. [18], MMs endorse this evaluation by outlining anticipated, typical, logical, and desired *evolution paths*. Furthermore, Mettler & Rohner [19] argue that, as formality is incorporated into the organizational development activities, decision makers are given a pragmatic instrument to determine whether potential benefits have been realized or not.

To delineate this evolution path, either a top-down or a bottom-up approach is conceivable [20]. With a top-down approach, a fixed number of maturity stages or levels is specified first and further corroborated with characteristics (typically in form of specific assessment items) that support the initial assumptions about the maturity distribution. When using a bottom-up approach, distinct characteristics or assessment items are determined first and clustered in a second step into maturity levels to induce a more general view of the different steps of the evolution path. In this sense, it is possible to define a different scale, number and type of maturity level for each of these clusters or *focus areas* [21].

Independent of whether a top-down or bottom-up approach is applied for describing the evolution path, reflections with respect to uncertainty of this evolution path have to be made [22]. In this regard, theories of emergence and diffusion of innovations reveal interesting foundations. According to Utterback & Abernathy [23],

[1] It is not the motivation of this paper to present an in-depth discourse on future BI-trends, but rather to show how the Rasch algorithm can be applied for maturity model construction.

the progress of a particular innovation follows an S-curve (left side of Fig. 1). Innovation (mainly originating from many minor product or process improvements) passes through different stages of maturity. Of particular interest is thereby the disruptive phase where a dominant design of a solution (i.e. a general agreed standard or best practice) becomes evident. However, dominant designs may not be better than other designs, but acceptance of an innovation will be on peak (right side of Fig. 1).

As regards the development of MMs, the cognition of the state in which an innovation is situated is thus extremely important, especially when the model is prescriptive. For instance, when focusing on the development of a MM for an emerging phenomenon, the levels of maturity may be extremely uncertain given that no dominant design is found already. Furthermore, only limited 'test cases' are available for effectively observing the accuracy of the recommended improvement actions, thus reducing the possibilities for attaining a reliable sample. Accordingly, this may have an effect on the appreciation of the MM as such, since the described evolution path may be perceived as speculation. On the other hand, when concentrating on a mature domain with lots of 'test cases' to rely upon, the levels of maturity may be clearer. However, the utility of the model may be reduced since the demand for guidance is lower. In such a case the results from an appraisal may be understood as 'bureaucracy' or 'platitude' as no substantial benefits may be gained.

A similar train of thoughts can be made when considering the diffusion of innovations. When using too fundamental or forward-looking criteria for the maturity assessment of an organization, the application of the model will show an accumulation of the results on a predefined sophistication level. For instance, Hayes & Zubrow [24] discovered that 73 percent of the assessed organizations between 1987 and 1994 were stuck on the first level of the Capability Maturity Model (CMM) because the requirements of the process area 'project management' were far too hard to meet. Therefore when defining the levels of maturity for a particular domain, a trade-off between the state of an innovation's uncertainty and its actual diffusion (which assists in predicting whether and how an innovation will be successful) has to be considered in order to guarantee 'useful insights' (i.e., trustworthy but not too obvious improvement activities) from the application of the model.

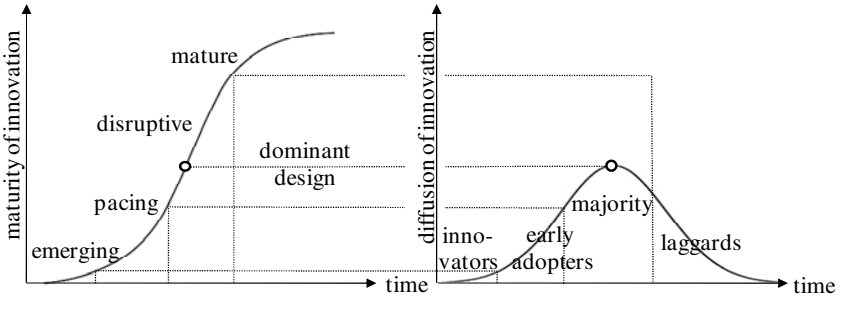


Fig. 1. Relationship between Maturity and Diffusion. Adapted from [22].

2.2 Maturity Model Development Methods

In contrast to other artifacts of the type "model", e.g. reference process models [25], only limited knowledge is available on how to systematically delineate the evolution path and how to instantiate the corresponding MM. As for now, five distinct development processes have been extensively discussed in current literature [18, 20-22, 26][2]. All of these processes share five generic design steps: (1) *Identify need or new opportunity*: Developing MMs by conducting design-oriented research means finding solution patterns for important unsolved problems or giving advice in solving problems in more effective or efficient ways [15]. According to Järvinen [27], a business need is not necessarily required but a new opportunity as "opportunity-based innovation can have a great economic value". As discussed in the prior section, the demand for a new MM strongly depends on the diffusion and maturity of the subject itself, given that mature themes typically require less explanation than emerging ones. (2) *Define scope*: In order to develop a useful model, the domain must be scoped properly [21]. This means that it has to be decided whether certain assumptions and characteristics are included or not. According to de Bruin et al. [20], this will "set the outer boundaries for model application and use". (3) *Design model*: The artifact as such is constructed in a third step. Becker et al. [18] and de Bruin et al. [20] both suggest a top-down approach by primarily defining a kind of grid or architecture of the relevant domain dimensions and sub-dimensions and 'filling' these dimensions with typical characteristics using focus groups/delphi method, creativity techniques, case studies, or literature reviews. Quantitative methods are less frequently used for constructing MMs. An example is the Rasch algorithm-based approach [22, 28], which is discussed and enhanced in this paper. (4) *Evaluate design*: Due to the fact that the acceptance of a MM critically depends on its utility, validity, reliability, and generalizability, evaluation is a crucial step in every design science research project [15]. (5) *Reflect evolution*: Finally, the MM has to be maintained and further development will be needed given that some model elements will get obsolete, new constructs will emerge, and assumptions on the different levels of maturity will be affirmed or refuted [18]. Therefore, even in an early stage it is important to also reflect on how to handle alterations in model design and deployment. However, this design step has attained little attention yet. Table 1 provides an overview of the applicable development methods and their adequacy for MM construction.

3 Using the Rasch Algorithm for MM Construction

3.1 Background and General Assumptions

Dekleva & Drehmer [29] favor the usage of the Item Response Theory (IRT) in the context of MMs. IRT has been developed to overcome deficiencies with classical test theory. Both theories aim to improve the reliability and validity of standardized tests or questionnaires by deriving conclusions about difficulty of items or the ability of participants [30]. IRT assumes that responses to questionnaire items are dependent on non-measureable respondent characteristics (a single latent variable or trait θ) and on item characteristics. Respondent characteristics could be psychological traits, but also

[2] For a more detailed analysis of the distinct development methods, please refer to [21, 27].

Table 1. Maturity Model Development Methods

Phase	Design method	Exemplary Source
1. Identify need or new opportunity	• Creativity techniques • Focus groups • Case studies • Literature review • Survey	• (Becker, et al. 2009) • (Mettler 2010a) • (de Bruin, et al. 2005) • (Van Steenbergen, et al. 2010)
2. Define scope	• Informed arguments • Scenarios	• (Van Steenbergen, et al. 2010) • (Mettler 2010a)
3. Design model	• Top-down: ▪ Delphi method ▪ Case studies ▪ Literature review • Bottom-up: ▪ *Algorithmic analysis* ▪ Informed arguments ▪ Ontologies	 • (Becker, et al. 2009) • (De Bruin et al. 2005) • (Van Steenbergen, et al. 2010) • (Mettler 2010a) • (Mettler 2010a)
4. Evaluate design	• Functional testing • Structural testing • Survey • Focus groups • Interviews	• (Mettler 2010a) • (Mettler 2010a)
5. Reflect evolution	• Field study • Interviews	

knowledge and specific capabilities. Determining a latent trait, IRT is a probabilistic alternative to factor analysis.

By basically counting properly answered questions, IRT calculates a score for the difficulty of items β_i and the ability of participants θ_v. Both scores are measured on the same interval scale. In the context of MM construction, the measurement of item difficulty supports the inductive allocation of items onto maturity levels and the capability of participants supports the assessment of organizations. The basic assumption and principle is schematically outlined in Fig. 2, where the responses of a dichotomous questionnaire are ordered in a response matrix. The sum scores are used to order the capabilities and difficulties.

Treating raw scores directly as measurements for the difficulty of items and capabilities of organizations, as in Fig. 2, does not yield in adequate results. Such a procedure tends to concentrate around middle scores and does not contrast the results of the more capable and less capable organizations, e.g. a very difficult item has the same contribution to the score as an easy item [30]. Therefore, the raw scores are first converted in their success-to-failure ratio, which allows displaying items and abilities ordered on one scale. Secondarily, the values are converted to their natural logarithm which calibrates the difficulty and ability around zero and therefore the score from a merely ordinal scale to interval scale, a log odds unit scale (see Table 2). A logit of zero corresponds to average difficulty (or ability). An additional logit doubles, a subtraction bisects difficulty or ability [31].

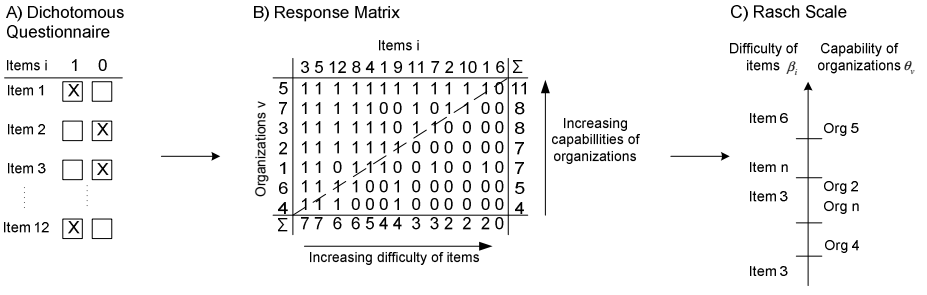


Fig. 2. Schematic Principle of the Rasch Algorithm

Furthermore, raw scores do not recognize patterns of responses by organizations. For example in Fig. 2, the raw score of organization 2 is orderly, as all easy items are properly answered and all difficult items are not. But it would be unfair to treat it similarly to organization 1 with the same raw score, but a pretty mixed response pattern; the same is true for the difficulty of item 10 compared to item 1. Therefore, using the Rasch algorithm, the capabilities and difficulties are tested against estimates based on a item response function (IRF) [32]. The IRF is based on the nonlinear monotone homogeneity model (MHM) which fits to the basic consideration of MMs in section 2.1. It states that highly skilled organizations have a high probability of having successfully implemented easy characteristics. Similar, low skilled organizations have a low probability of using advanced practices. The IRF is defined as the conditional probability of success $f_i(\theta_v)$ for an organization v to solve item i. Consequently, the calculation of estimates of difficulty and capabilities is based on the inverse IRF using Maximum Likelihood Estimates (MLE) [32].

The degree of difficulty of an item, the ability of organizations θ_v, the IRF, and estimates can be calculated by the following steps presented in Table 2 [33].

Table 2. Calculation of the Measures, IRF and Estimates

Step	Formula	Description
0	-	Eliminate all items, which have been solved by all or none of the organizations, as no additional information is provided.
1	$\beta_i = \ln\frac{1-p}{p}$;	Calculate degree of difficulty of an item *i*: share of not or incorrect answered items (1-*p*) divided by share of correct answered items (*p*).
2	$\theta_v = \ln\frac{p}{1-p}$;	Calculate ability of an organization v: share of correct answered items (*p*) divided by share of not or incorrect answered items (1-*p*).
3	$f_i(\theta_v) = P(X_{vi} = 1\|\theta_v, \beta_i) = \frac{\exp(\theta_v - \beta_i)}{1 + \exp(\theta_v - \beta_i)}$	Calculate the conditional probability of success $f_i(\theta_v)$. Using one scale, it is based on the difference between the ability of organizations θ_v and the difficulty of the items β_i, so that for a given value θ_v the probability decreases with increasing difficulty β_i.
4	$\hat{\theta}_v, \hat{\pi}_{r_v i}$	The estimates of difficulties and items are derived by using MLE based on the inverse IRF.

The Rasch algorithm is able to handle binary scales and rating scales [30]. For simplicity reasons, the paper provides a short introduction using binary scales first.

Key indices for testing are OUTFIT and INFIT [32]. Both are based on chi-square fit statistics using the residuals between the observed and the expected value $\hat{\pi}_{r_v i}$ under the IRF (see Table 3). They indicate how accurately or predictably data fits the MHM.

Table 3. Calculation of OUTFIT and INFIT indices

Formulary	Description	Expectation
$OUTFIT_i = \frac{1}{V}\sum_{v=1}^{V}\frac{(x_{vi}-\hat{\pi}_{r_v i})^2}{\hat{\pi}_{vi}(1-\hat{\pi}_{vi})}$	OUTFIT is based on the conventional sum of squared standardized residuals. It is more sensitive to extreme scores.	Expectation of 1, values substantially below 1 indicate dependency in data; values substantially above 1 indicate noise.
$INFIT_i = \frac{\sum_{v=1}^{V}(x_{vi}-\hat{\pi}_{r_v i})^2}{\sum_{v=1}^{V}\hat{\pi}_{vi}(1-\hat{\pi}_{vi})}$	INFIT is an information-weighted sum. The statistical information is its variance. This is larger for targeted observations, and smaller for extreme observations. So it is inlier-sensitive.	Expectation of 1, values substantially below 1 indicate dependency in data; values substantially above 1 indicate noise.

For details and mathematical specifications of the underlying assumptions, the specialized Rasch literature is recommended [30-32, 34].

3.2 Rasch Algorithm-Based Maturity Model Construction

The MHM and the positioning of difficulty of items and ability of organizations fit to the basis considerations of MMs, which has already led to applications of Rasch for MM construction by selected researchers [22, 28]. To make it applicable for MM construction in the field of IS research, the basic model needs three modifications, without challenging the basic assumption of the Rasch algorithm:

First, in the context of complex socio technical systems the expressive power of rating scales, e.g. a Likert scale from one to five, is superior to dichotomous scales. A Likert scale clearly recognizes that the questions are requiring merely expressed opinions than just simple right or wrong answers [35]. Therefore, rating scales instead of dichotomous scales should be deployed. The Rasch algorithm is already prepared for the handling of rating scales. But the determination of the characteristics is slightly more complex, for details refer to the specialized Rasch literature [30, 34].

Second, in order to develop MMs in the context of IS, researchers should not only ask for the actual situation of an item i at organization v A_{vi}, but also for the desired situation of item i at organization v D_{vi}. Following the principle of economic efficiency [36], the overall utility function of an item i at organization v is not necessary monotonically increasing, but could be limited by D_{vi} providing an upper bound for the item. The potential and desired improvement is then provided by the delta value between desired and actual values $X_{vi} = D_{vi} - A_{vi}$. A negative delta expresses overcompliance or undesired developments. The delta value X_{vi} can now

serve as the input variable for the Rasch algorithm. Considering that MMs should provide an individual as-is assessment, but a common and consistent development perspective, this paper proposes the usage of a modified delta between the individual actual values A_{vi} (organization specific) and common desired values $\widetilde{D}_i$ (median across all organizations) $X_{vi} = \widetilde{D}_i - A_{vi}$. The median $\widetilde{D}_i$ is able to express the importance of one item for all organizations. Compared to the arithmetic mean, the median is much more stable for extreme values [30]. Thus, the median is preferred over the mean. The difficulty of an item per organization is then given by the delta value: A high positive gap expresses a difficult and desired item. Negative gaps and also values on the threshold itself express easy items. Following common practice of Rasch model application, categories are collapsed with respect to the situation at hand [30]. Therefore, all negative gaps are collapsed into one category, as they all express overcompliance. Also very large gaps are collapse into one category as there is hardly any distinction between "very, very" difficult items and "very, very, very" difficult items. Therefore, a re-coding of data is applied resulting in five values. The result is presented in Table 4. Following Rasch model application for Likert scales, more easy items are expressed with a higher value.

Table 4. Coding of Actual and Desired Values

Delta Δ	<0	0	1	2	>2
X_{vi}	5	4	3	2	1
Interpretation	Easy items			Difficult items	

Third, the Rasch algorithm only yields a single ordinal scale that represents the logit measure of each item and organization, but not distinct maturity levels. In order to overcome subjectivity in defining maturity levels (e.g. by defining level one items to have logits smaller than -1), we propose the usage of cluster analysis on the basis of the item logits. The purpose of clustering is to investigate "a set of objects in order to establish whether or not they fall […] into groups […] of objects with the property that objects in the same group are similar to one another and different from objects in other groups." [37] Within various clustering methods, agglomerative algorithms have the largest significance in practice and are selected here for maturity level distinction [38]. Based on the measure of the items, they start with n clusters, each containing a single item. Using a standardized distance measure, step by step the number of clusters is reduced. As most MMs use five maturity levels [8, 11], the anticipated number of clusters is set to five.

4 Evaluation of the Method

In order to show the applicability of the method proposed, we outline a design exemplar for a BI MM. Section 4.1 provides an overview of the survey design and sample. Section 4.2 presents the results of the Rasch algorithm and the cluster analysis. Based on these results, section 4.3 reflects on the method presented at hand.

4.1 Survey Design and Data Collection

For the design exemplar, a questionnaire is used that is structured into four dimensions. Following well accepted management and IS frameworks, three functional BI dimensions of "strategy" ("what"), the "organization and processes" ("how") and the "IT support" ("with what") are differentiated [39]. Additionally, the fourth dimension "quality of service" focuses on how well the BI organization performs in its environments. Therefore, it contains questions related to non-functional aspects such as user satisfaction, performance and scalability [40]. For each dimension, two up to six questions are defined, resulting in a total of 14 questions. Each question contains two up to six characteristics, which are assumed to have different levels of difficulty, respectively maturity. This results in a sum of 58 items. A paper-based questionnaire is used to collect data. The responses are scored on a five-point Likert scale. The participants of an international conference for practitioners in the field of BI held in 2010 in Switzerland served as sample. From the 144 participants of the conference, 51 returned the questionnaire yielding in a response rate of 35.4%. In Table 5, the characteristics of the sample are outlined.

Table 5. Sample Characteristics

Industry sector	No	%	Employees	No	%
Automotive	2	4	0-100	9	18
Chemical & pharmacy	3	6	100-1000	11	22
Services	13	25	1000-10000	13	25
Utilities	3	6	> 10000	13	25
Finance & banking	8	16	Not available	5	10
Healthcare	1	2	Sum	51	100
Wholesale & retailing	4	8	Position of respondent	No	%
Techn., new media & telecom.	5	10	Business organization	5	10
Transportation & logistics	2	4	IT organization	29	57
Others	7	14	Mixed org. unit	15	29
Not available	3	6	Not available	2	4
Sum	51	100	Sum	51	100

4.2 Results

The BIGSTEPS software 2.82 [32] has been used to obtain item calibrations. Important output statistics are the measure of difficulty (the logit values), the standard error, and the fit statistics of INFIT and OUTFIT for each item. Table 6 contains the results ordered by descending measure, i.e. difficult items have a higher measure than easy items. The fit values are all around 1 and therefore satisfying the expectation (see Table 3). The result is a flat list of ordered items. Although first conclusions can be drawn from such a list, the measures do not allow a division in distinct maturity levels. For clustering, we use the hierarchical cluster analysis of SPSS 19 (squared Euclidean distance, Ward's method) and set the desired number of maturity levels to five. The results of the cluster analysis are also exhibited in Table 6.

Table 6. Results of Rasch Algorithm

Logit	Error	Infit	Outfit	Item	Cl.	Short description
0.91	0.15	0.61	0.62	4.1.d	5	Proactive data quality management
0.79	0.14	1.26	1.23	1.3.c	5	Systematic and comprehensive measurement of real BI usage
0.72	0.14	1.2	1.18	1.3.b	5	Balanced Scorecard incl. quality, cost and user satisfaction
0.56	0.15	1.25	1.19	1.1.e	4	BI steering committee within business
0.56	0.14	1.02	1.01	2.2.d	4	Balanced mix of central and decentral organizational units
0.5	0.14	0.96	0.95	1.3.e	4	Portfolio mgt. for systematic BI roadmap
0.45	0.14	0.81	0.81	1.2.c	4	Regular update on BI strategy
0.39	0.14	0.59	0.61	1.3.d	4	Value-oriented dev. of BI, e.g. using business cases
0.37	0.13	0.76	0.84	2.3.b	4	Defined governance & standards for content
0.37	0.13	1	1.03	2.4.b	4	Dev. of BI based on standardized BI processes
0.37	0.13	0.62	0.62	2.6.b	4	Use of BI by middle-management
0.37	0.14	1.43	1.42	3.1.c.	4	Flexible, proactive analytics
0.33	0.13	1.06	1.05	1.2.b	4	BI strategy with focus on organization, processes as well as technology and tools
0.31	0.14	1.06	1.07	2.5.b	4	Central BI operations based on ITIL
0.24	0.14	0.85	0.86	4.1.a	3	Defined and documented roles for data quality mgt.
0.21	0.14	0.96	0.94	4.1.e	3	Std. definitions for key perf. indicators
0.2	0.14	1.13	1.14	4.2.e	3	BI operations based on well defined service-level-agreements (SLAs)
0.18	0.14	0.76	0.8	2.3.c	3	Defined governance & standards for management
0.17	0.14	0.47	0.48	4.2.d	3	Performance is satisfying for users
0.14	0.14	0.7	0.72	4.2.c	3	Timeliness: Usage of up-to-date tools and frontends
0.11	0.14	1.43	1.42	1.1.b	3	Multitude of decentral sponsors from business
0.11	0.14	0.91	0.9	4.1.b	3	Defined processes. for data quality mgt.
0.11	0.14	0.65	0.64	4.2.f	3	Cost efficient BI operations
0.09	0.13	1.29	1.36	2.1.d	3	Business partner - consulting of business lines
0.08	0.13	0.87	0.85	2.6.c	3	Use of BI by specialist
0.06	0.13	1.28	1.27	1.2.a	3	BI strategy with focus on techn. and tools
0.05	0.13	1.22	1.23	1.3.a	3	Std. cost and profit calculation of BI
0.02	0.14	0.82	0.83	1.1.d	3	Central influencing sponsor from business
0.01	0.13	1.09	1.08	3.1.d	3	Integration of different frontends, using drill-through from standard reports into OLAP cubes
0	0.14	0.81	0.8	4.2.b	3	Homogeneity: Usage of a few and coherent BI tools
-0.01	0.14	1.56	1.58	3.2.b	3	Partial integration in global systems (e.g. finance data warehouse)

Table 6. (*continued*)

-0.01	0.14	0.96	0.98	3.2.e	3	Balanced mix of central and decentral systems based on organizational structure
-0.03	0.13	1.25	1.24	2.6.d	3	Operational usage of BI
-0.06	0.14	0.91	0.91	1.1.f	3	BI steering committee within IT
-0.06	0.14	0.96	0.96	2.4.c	3	Development of BI using agile development methods (e.g. SCRUM)
-0.08	0.14	0.9	0.9	4.1.c	3	Most important data objects are defined for whole enterprise
-0.08	0.14	0.97	0.95	4.1.f	3	Standardized definitions for master data
-0.1	0.14	0.64	0.63	3.1.a	3	Static reports
-0.11	0.14	0.86	0.86	2.1.a	3	Defined governance & standards for dev.
-0.11	0.13	0.83	0.83	2.3.d	3	Operator of infrastructure
-0.15	0.15	1.02	1.02	2.4.d	2	Hybrid development of BI mixing agile development and waterfall methods
-0.22	0.14	1.2	1.21	2.1.c	2	Provision of standardized services
-0.27	0.14	1.31	1.3	2.2.c	2	Central BI organization
-0.28	0.14	1.38	1.38	3.2.d	2	Highly central data warehouse
-0.28	0.14	0.97	0.96	4.2.a	2	High availability: No breakdowns, maintenance in well defined and short time slots
-0.3	0.14	1.24	1.23	2.2.b	2	Decentral BI org. with central CIO org.
-0.31	0.14	1.39	1.44	2.6.a	2	Use of BI by top-management
-0.32	0.14	0.73	0.73	2.3.a	2	Defined governance & standards for tools and applications
-0.33	0.14	0.76	0.75	2.3.e	2	Defined gov. & standards for operations
-0.39	0.14	0.91	0.93	2.4.a	2	Dev. of BI based on std. IT processes
-0.45	0.15	1.19	1.21	1.1.c	2	Central influencing sponsor from IT
-0.49	0.15	1.62	1.59	2.1.b	2	Project oriented development
-0.55	0.16	0.72	0.79	3.2.f	2	Decentral, but harmonized systems (e.g. standardized master data)
-0.58	0.15	1.16	1.13	3.1.b	2	Ad-hoc analyses (OLAP)
-0.65	0.15	1.2	1.12	2.5.a	2	Central operations of BI
-0.81	0.16	0.88	0.92	1.1.a	1	Multitude of decentral sponsors from IT
-0.89	0.16	1.15	1.14	2.2.a	1	Decentral BI org. and responsibilities
-0.91	0.17	0.98	1.08	3.2.c	1	Decentralized warehouses and central enterprise warehouse

Based on the results of the application of the Rasch algorithm and the cluster analysis, an initial MM can be derived. Fig. 3 presents a sketch of the MM, based on the meta structure of strategy, organization & processes, IT and quality of service as outlined in the questionnaire structure.

The results can be interpreted as follows: Level 1 is populated by capabilities reflecting an early stage of decentralized BI organization. Level 2 emphasizes stronger orientation towards centrally managed BI in terms of governance and organization. Furthermore, services and processes are much more standardized on level 2. This could be interpreted as a first wave of optimized BI operations. On level 3, there is a strong focus on governance, strategy, and development. In addition, the BI infrastructure is also more balanced between centralized and decentralized

		Level 1		Level 2		Level 3		Level 4		Level 5	
		Item	Description	Item	Description	Item	Description	Item	Description	Item	Description
1. Strategy	1.1 Governance	1.1.a	Multitude of decentral sponsors from IT	1.1.c	Central influencing sponsor from IT	1.1.b 1.1.d 1.1.f	Multitude of decentral sponsors from business Central influencing sponsor from business BI steering committee within IT	1.1.e	BI steering committee within business		
	1.2 Strategy					1.2.a	BI strategy with focus on technology and tools	1.2.b 1.2.c	BI strategy with focus on organization, processes as well as technology and tools Regular update on BI strategy		
	1.3 Controls					1.3.a	Standardized cost and profit calculation of BI	1.3.d 1.3.e	Value-oriented development of BI, e.g. using business cases Portfolio management for systematic BI roadmap	1.3.b 1.3.c	Balanced scorecard incl. quality, cost, and user satisfaction Systematic and comprehensive measurement of real BI usage
2. Processes & organization	2.1 Role			2.1.b 2.1.c	Project oriented development Provision of standardized services	2.1.a 2.1.d	Operator of infrastructure Business partner - consulting of business lines				
	2.2. Organization	2.2.a	Decentral BI organization and responsibilities	2.2.b 2.2.c	Decentral BI organization with central CIO organization Central BI organization			2.2.d	Balanced mix of central and decentral organizational units		
	2.3. Governance processes			2.3.a 2.3.e	Defined governance & standards for tools and applications Defined governance & standards for operations	2.3.c 2.3.d	Defined governance & standards for management Defined governance & standards for development	2.3.b	Defined governance & standards for content		
	2.4 Change			2.4.a	Development of BI based on standardized IT processes	2.4.c 2.4.d	Development of BI using agile development methods (e.g. SCRUM) Hybrid development of BI mixing agile development and waterfall methods	2.4.b	Development of BI based on standardized BI processes		
	2.5 Run			2.5.a	Central operations of BI			2.5.b	Central operation of BI based on ITIL		
	2.6 Usage			2.6.a	Use of BI by top-management	2.6.c 2.6.d	Use of BI by specialists Operational usage of BI	2.6.b	Use of BI by middle-management		
3. IT	3.1 Frontend			3.1.b	Ad-hoc analyses (OLAP)	3.1.a 3.1.d	Static reports Integration of different frontends, e.g. drill-through from standard reports into OLAP cubes	3.1.c	Flexible, proactive analytics		
	3.2 Backend	3.2.c	Decentralized warehouses and central enterprise warehouse	3.2. d 3.2.f	Highly central data warehouse Decentral, but harmonized systems (e.g. standardized master data)	3.2. b 3.2.e	Partial integration of information in global systems (e.g. finance data warehouse) Balanced mix of central and decentral systems based on organizational structure				
4. Quality of service	4.1 Data quality					4.1.a 4.1.c 4.1.b 4.1.f 4.1. e	Defined and documented roles for data quality management Most important data objects are defined for the whole enterprise Defined processes for data quality management Standardized definitions for master data Standardized definitions for key performance indicators			4.1.d	Proactive data quality management
	4.2. System quality			4.2.a	High availability: No breakdowns, maintenance in defined and short time slots	4.2. b 4.2.c 4.2.d 4.2.e 4.2.f	Homogeneity: Usage of few and coherent BI tools Timeliness: Usage of up-to-date tools and frontends Performance is satisfying for users BI operations based on well defined service-level-agreements (SLAs) Cost efficient BI operations				

Fig. 3. Algorithm based Maturity Model

operations. BI applications are used by a broader scope of users on this level. Moreover, data quality and system quality have improved. This indicates a widespread diffusion of BI development and BI usage throughout the company. Level 4 is dominated by restructuring, value orientation and further optimization of BI. With growing BI experience, companies consolidate and streamline their widespread BI activities. The capabilities on level 5 are focused on the proactive and fact-based management of ongoing BI operations.

4.3 Discussion

The application of the proposed methodology yields in an initial version of a BI MM. It clearly follows an S-curve or normal distribution, with few capabilities at the beginning, an increasing number of capabilities on levels 2 and 3, and fewer additional activities on levels 4 and 5. A comparable development has also been proposed within the data warehouse MM by Watson et al. [41] and the TDWI BI MM [42]. It is also a basic assumption of MM construction (see section. 2.1). Furthermore, not only the number of allocations, but also the interpretation from a content perspective leads to a consistent and reasonable initial version of a BI MM. However, it should be clearly noted that such an initial MM needs additional iterations as proposed by the overall design science research approach [14, 15] as also for MM construction [18]; e.g. by leveraging focus groups for level and item discussion. Finally, it can be concluded that in the context of this BI-related analysis the proposed method has been demonstrated to be a useful means for empirically grounded and therefore more rigorous MM construction.

5 Conclusion and Outlook

MMs have become an established means in the IS community to systematically document and guide the development of organizations using archetypal capability levels. Despite its popularity, the concept has been criticized as lacking a generally accepted design methodology and formal basis, leading to a widespread development of MMs of disputable quality. This paper presents a statistical, psychometric-driven approach for designing MMs based on the Rasch algorithm and cluster analysis. Compared to previous applications of the Rasch algorithm in this field [22, 28, 29], the proposed method takes the current situation as well as corresponding targets into account. Moreover, the arbitrary assignment of items to maturity levels is overcome by applying a clustering algorithm.

The applicability of the method is limited to mature domains as there is a relative large sample needed for identifying items and derive maturity levels. This is also accompanied with larger efforts compared to other MM construction methods. Even though the result is more rigorous, it could lead to counterintuitive results during the first iterations and therefore may be lacking acceptance compared to a top-down development. Future research should be directed towards integrating the method into a full MM design process, i.e. using the Rasch algorithm for holistic organizational assessment. The algorithm could also be used for situational MM design (e.g. by using specific sample sets) and MM evolution on the basis of iterative survey execution.

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